

Hybrid Image Segmentation for Retinal Cyst Detection in OCT Images

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INTRODUCTION

Retinal Cyst are fluid filled hypo-reflective lesions that are the key biomarkers in eye diseases such as diabetic macular oedema (DME) and other age-related macular degeneration (AMD). OCT image (Optical Coherence Tomography) is a gold standard for imaging modalities and visualising retinal microstructures with micrometre level precision. Accurate segmentation and detection of this cyst is crucial for diagnosis, treatment planning, and long-term monitoring. Manually delineation is labour intensive, time consuming and suffers from inter grader variability, and is impractical for high-volume screening.

Classical image segmentation approaches remain important where computational efficiency, explainability, and limited data prevent the use of deep neural networks. Image segmentation for cyst detection is particularly challenging due to low contrast, noise, heterogeneous cyst shapes, and the presence of layered retinal structures that are confused with the cyst.

There are several prior works that have studied classical cyst segmentation.[1] uses region flooding and texture analysis to segment dark fluid pockets.[2] analyses cyst using local intensity features and morphological rules.[3] explores graph-based and region-based segmentation on OCT layers for cyst segmentation. The key parameters which this project analyses from these works: (1) cyst are typically dark and smooth interiors, (2) retinal Structure introduces strong horizontal gradients, and (3) combining multiple cues like – intensity, texture, and geometry - improves robustness.

Driven by these findings and taking the guidance from the lecture and practical contents from *Segmentation Fundamentals* (L9) and *Segmentation Algorithms* (L10), this project develops a classical hybrid segmentation system combining region flooding, robust contrast enhancement, entropy – based textured filtering and the watershed refinement. This algorithm is constructed in phases, starting from simple baselines and iteratively improving towards a refined model. The final implementation achieves consistent cyst segmentation on the provided dataset using only classical methods, satisfying the requirements of this assignment.

Methodology

This project proposes a system which follows a carefully engineered ,multi-stage classical segmentation pipeline .The design of each phase is strongly influenced by the literature on cyst segmentation in OCT Images – [1],[2],and [3] and concepts from the lectures .By iterative segmentation and experimentation , several intermediate approaches like K-means , global thresholding , raw watershed were tested and before coming to the final approach of the project , with a hybrid strategy.

Pipeline overview

The final algorithm , used for this project works through the following steps :

1. Pre-processing
2. Retina Region Extraction (region of interest)
3. Cyst Enhancement
4. Strict cyst extraction- phase 4
5. Entropy – Guided Flooding and Watershed Refinement – phase 5
6. Evaluation and Visualization

Pre-processing

Before segmentation, each OCT image is pre-processed to reduce unwanted information-noise and enhance local contrast while preserving meaningful information and the retinal structure.

- 1 Intensity Normalisation
The raw image is converted to float32 and normalized to the range of 0-255:

$$I_{\text{norm}} = 255 \cdot \frac{I - \min(I)}{\max(I) - \min(I) + \epsilon}$$

This step is essential for consistency across OCT slices, and particularly because Contrast and thresholding functions assume values within a controlled range.

- 2 Median Filtering
Median filtering removes salt and pepper noise from the oct image and help for consistent result. Implemented it as introduce in lab9 workbook. This filter is effective in removing speckle noise, commonly present in OCT
- 3 Gaussian Smoothing ($\sigma = 1.0$)
A Gaussian filter Smoots homogeneous cyst region while preserving retinal structure. This step aligns with the smoothing for pre-processing is used by Fabritius et al. [2] to stabilise texture descriptors.

Retina Region Extraction (Region of Interest)

As the cyst only appears within the retina layer, extracting the retain region is essential to prevent false positives and reduce computation.

This step combines three ideas:

- 1 Otsu Thresholding: For coarse tissue segmentation
- 2 Canny Edge Detection: to analyse retina region.
- 3 Anatomical band (rows 80-300): derived through dataset analyses and sported by González et al. [1] , who emphasised on restricting segmentation to plausible retina region.
- 4 Hole Filling
- 5 Mask Application

Otsu Thresholding

Otsu's method finds a global thresholding that separate foreground from background, it is not so accurate to detect the cyst, it reliably distinguishes retina from surrounding noise.

Canny Edge Detection

Edges reinforced transition between retina layer .Weak thresholds (0.1,0.3) ensure gradients inside the retina and preserved.

Anatomical band

The mask in refined by and-ing with affixed vertical band.

$$Mask = OtsuMask \cap CannyEdges \cap [80:300 \text{ rows}]$$

Hole Filling

Binary fill on holes create a coherent closed retina region suitable for region-based segmentation.

Mask Application

Everything outside the retina region is zeroed out. This mirrors the region-limiting preprocessing described by Chiu et al. [3].

Cyst Enhancement

Cyst appears as dark, low -contrast, hypo-reflective sacks. the enhance its separability a local enhancement step is applied.

- 1 Percentile-Based Contrast Stretching (1%–99%)
Inside the ROI (region of interest) the retina this which we got from the previous step, intensities are stretched using:

$$I' = rescale_{intensity}(I_{roi,p2,p98})$$

This normalises local contrast without amplifying noise.

- 2 Light Gaussian Smoothing ($\sigma = 0.6$)
This step further smooths out minor irregularities and noises present, to prepare it for the next step. This step aligns with smoothing strategies in region-flooding literature [1].

Strict Cyst Extraction

This is the backbone of this project algorithm, producing consistently accurate outputs.

These are it constituting steps:

CLAHE (adaptive histogram equalisation)

Contrast limited adaptive Histogram Equalization (CLAHE) is applied based on the retina ROI based on the effectiveness of local histogram equalisation in OCT Segmentation.

The equalization is calculated using:

$$I_{clahe} = equalize_{adapthist}\left(\frac{I}{255}, clip_{limit}=0.02\right) \cdot 255$$

Darkest 12% Pixel Selection

González et al. [1] observed that cyst belong to the lowest intensity percentile of the retinal structure. Following this, we compute:

$$p_{dark} = percentile(I_{clahe}, 12)$$

Pixels Darker than this threshold and inside the retina ROI are treated as cyst candidates.

Local Adaptive Thresholding

After repeated trials and visually evaluating, a block -size of 35 with offset of 0.03 was found optimal:

$$I_{bin}(x, y) = I_{clahe}(x, y) < T_{local}(x, y)$$

This ensures adaptation local variation in their visuals.

Morphological Refinement

- Opening disk (disk=2) removes noise.
- Closing disk (disk=3) seals gaps.
- Removes small objects (man_size=20) eliminates tiny false positives.

Entropy-Guided Watershed Refinement

This step enhances the output image from the previous phase and improves accuracy by combing texture and region-flooding principles.

Entropy Filtering

Cyst have low texture, while retina layers have high entropy to it microstructures. Rank entropy is computed using a disk of radius 3:

$$E(x, y) = \text{entropy}(I, \text{disk}(3))$$

A threshold of 5.5 is step in this model based on visual analyses and experimentation.

Watershed Segmentation

Watershed Segmentation is applied:

$$\text{watershed}(-D, \text{markers}, \text{mask} = \text{entropyMask})$$

This creates region boundaries closely aligning with the cyst shapes.

Morphological Constraints

After watershed, these constraints follow the method similar to Fabritius et al. [2] for discarding elongated regions or layer of variations within the retina.

- Opening (disk=2)
- Closing (disk=3)
- Remove small regions (<15 pixels).
- Shape constraints eccentricity < 0.98, extent < 0.95)

Output

$$\text{refined_mask (final cyst segmentation)}$$

Final Evaluation and Visualisation

Dice Coefficient

$$\text{Dice} = \frac{2 | P \cap G |}{| P | + | G |}$$

Intersection over Union (IoU)

$$\text{IoU} = \frac{| P \cap G |}{| P \cup G |}$$

Mean Intersection over Union (mIoU)

$$mIoU = \frac{1}{N} \sum_{i=1}^N IoU_i$$

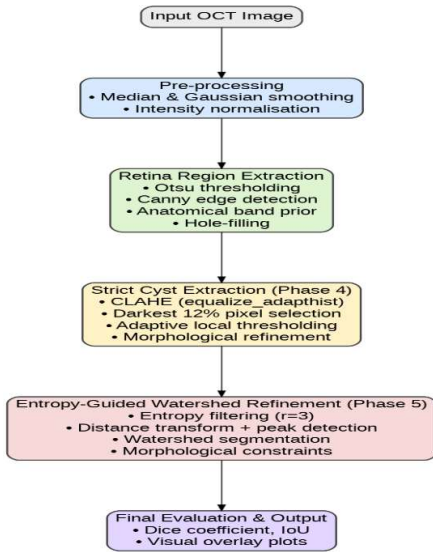


Figure 1 End-to-end pipeline for the proposed classical hybrid retinal cyst segmentation system.

Implementation Details

Tools & Libraries

- Python & Google Colab Notebook
- scikit-image (filters, morphology, watershed)
- scipy.ndimage (distance transforms, median filter)
- NumPy
- Matplotlib for visualisation

This proposed cyst segmentation system is implemented entirely in python using Google Colab .The core Libraries used is scikit-image ,SciPy, NumPy, and Matplotlib.scikit-image provided most classical computer -vision methods required for assignment ,including adaptive thresholding ,morphological operations ,entropy, filtering and watershed segmentation, all of which align to the technique used in lecture 9 and 10 .SciPy was also used for median filter. NumPy was used for metric calculation and Matplotlib was used to generate qualitative visualisations and overlay plots.

This combination is chosen to satisfy the course work requirements and tasks, using classical image processing algorithm rather than CNN-based methods.

Results

Quantitative Evaluation

Performance was evaluated on al the ten images given in the data set using Dice Coefficient and Intersection over union (IoU). These matrices quantify the overlap between predicted mask and the ground truth annotation images given in the data set, as specified in the assignment requirements.

Table 1 Quantitative analysis of segmentation performance across the 10 OCT training images

Image	Dice Coefficient	Intersection over Union
TRAINING1	0.3831	0.2370
TRAINING2	0.5771	0.4056
TRAINING3	0.0557	0.0286
TRAINING4	0.5983	0.4268
TRAINING5	0.5398	0.3697
TRAINING6	0.5208	0.3521
TRAINING7	0.1519	0.0822
TRAINING8	0.5737	0.4022
TRAINING9	0.2690	0.1554
TRAINING10	0.4846	0.3198
Average Dice: 0.4154		
Mean IoU (mIoU) :0.2779		

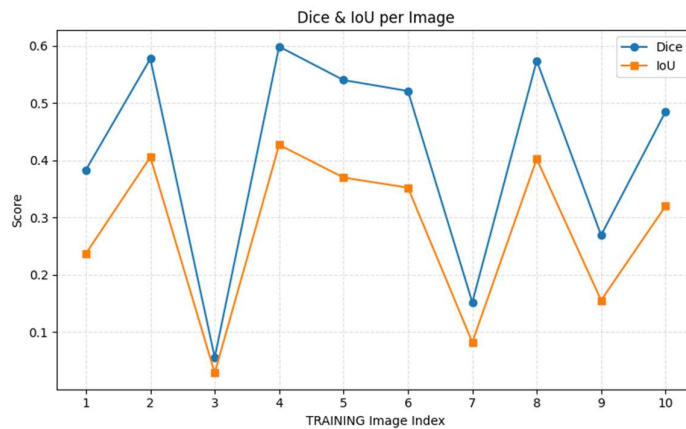


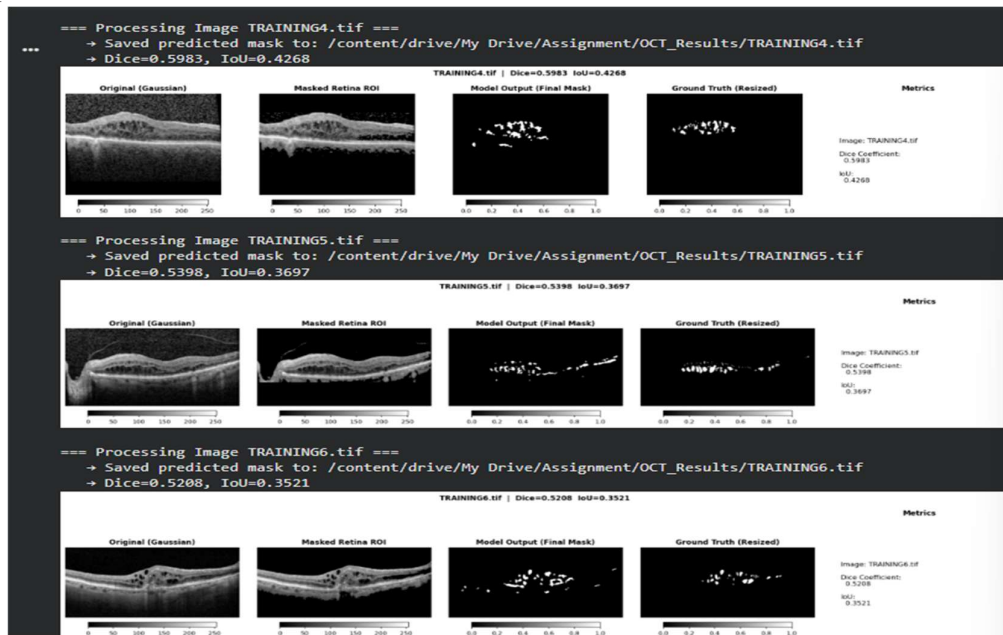
Figure 2 Dice and IoU scores for each image in the OCT training dataset.

Qualitative Visualisation

The processed OCT segmented is visualised in five column layouts comprising.

- 1 Original
- 2 Retina-masked
- 3 Predicted refined mask.
- 4 Provided Ground truth Image.
- 5 Dice and IOU value of each.

Output



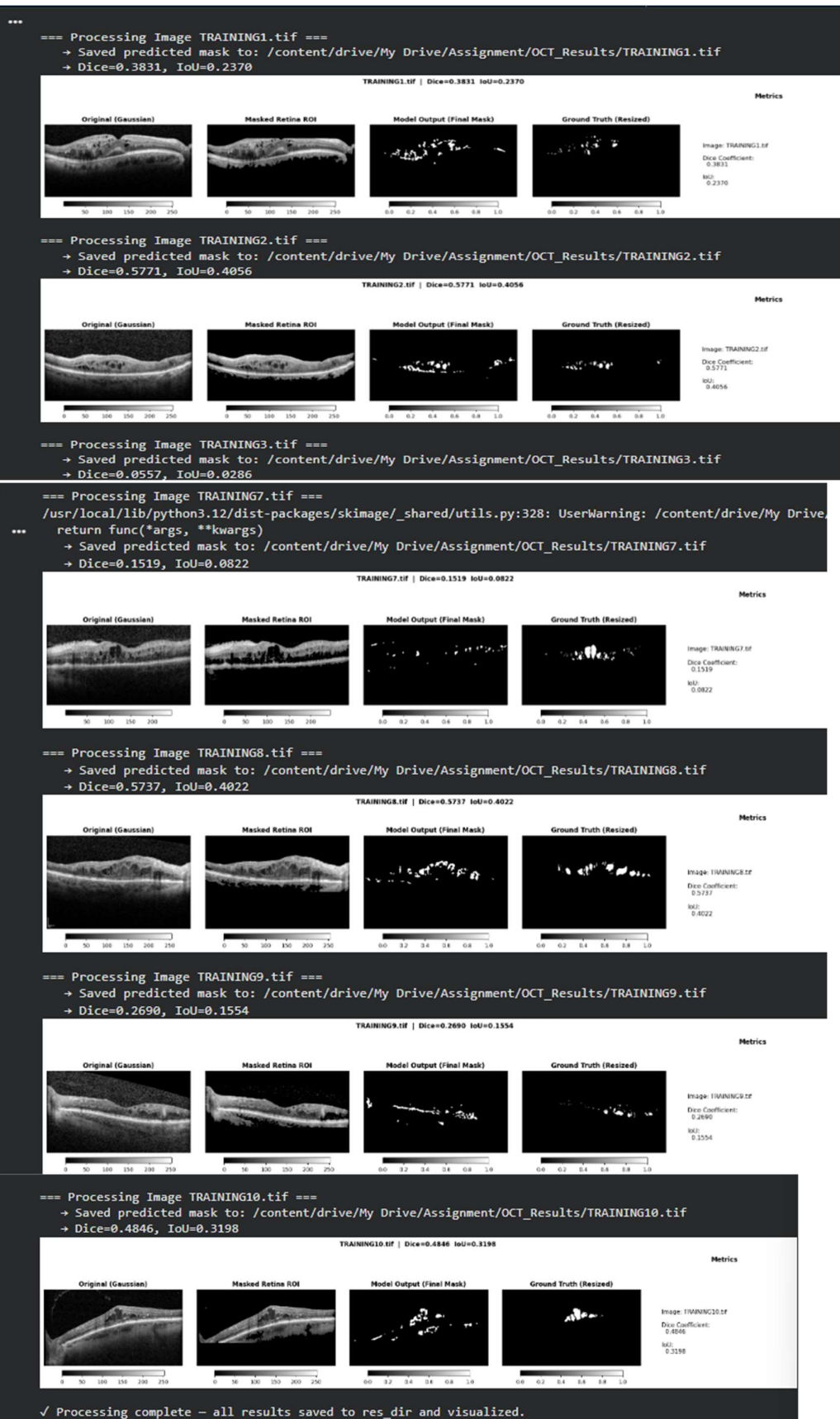


Figure 3 Screenshot of the output.

Discussion

Strengths

This processed hybrid segmentation method demonstrated strong robustness by combining intensity, texture analysis, and morphological cues -- avoiding weakness of single-strategy approaches. This approach with guided entropy watershed, effectively separates merged cyst cluster, while darkest - 12% selection ensures high precision in identifying hyper reflective regions of the cyst. This system is transparent, light weight, and full reproducible, meeting the assignment requirements and reflecting segmentation strategies commonly reported and used in literature.

Limitations

The thresholds like entropy <5.5 and darkest 12% were optimised for this data set to give the best result, which might not give the same result if using different Oct scans from outside the data set or from different machine. Hence it cannot be generalised. Watershed segmentation is also known to be sensitive to small variation in texture, which contributed to lower score in images a bit faint or have irregular cyst boundaries.

Comparison with Literature

The method aligns closely with González et al. [1], who similarly combined contrast enhancement and texture-based refinement. The use of entropy is consistent with texture driven approaches in Fabritius et al. [2]. Watershed based splitting resembles the cyst boundary, detection and enhancement strategy as described in Chiu et al. [3]. This This hybrid approach in this project helps in controlling over segmentation and improves cyst isolation unlike if we would just use region flooding only methods.

Conclusions

This project has successfully developed a fully classical retinal cyst segmentation pipeline for OCT scan. This Hybrid approaches – incorporated CLAHE enhancement, percentile based dark spot detection, retina region detection, entropy filtering and watershed refinement – is able to achieve a meaningful segmentation performance without the use of CNNs. This pipeline demonstrates explainability, reproducibility and alignment with both the academic literature and lecture content.

Future improvements may include adaptive entropy thresholding, learning based thresholding selection and extending to 3-D volumetric segmentation of the OCT data.

References

- [1] A. González et al., *Automatic Cyst Detection in OCT Retinal Images Combining Region Flooding and Texture Analysis*, 2017.
- [2] T. Fabritius et al., *Segmentation of Cystoid Macular Edema in Optical Coherence Tomography*, 2009.
- [3] S. J. Chiu et al., *Automatic Segmentation of the Choroid in OCT Images*, 2012.